AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 7

Lab Title: 7-segment display using BCD to 7-segment decoder 7447.

Student Name: Muammad Waqas **Reg. No:** 240756

Objective: To experimentally check the operation of 7-segment display using BCD to 7-segment decoder 7447.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

EXPERIMENT 07

Seven Segment Display

Objectives:

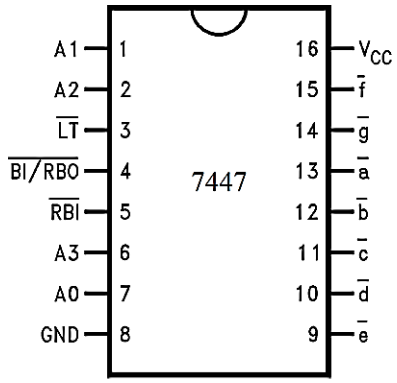
- To experimentally check the operation of 7-segment display using BCD to 7-segment decoder 7447.

Equipment required:

- ICs as required
- Digital Electronics Trainer

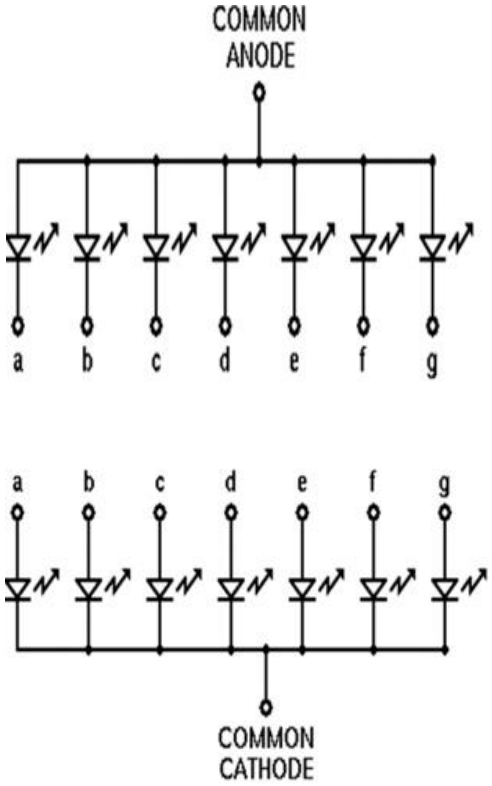
Background Knowledge:

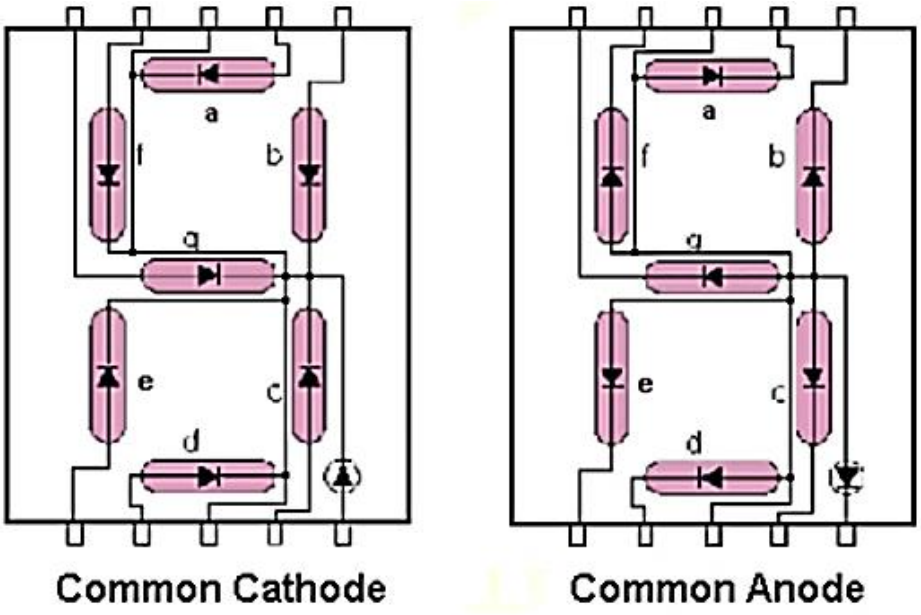
A BCD to seven-segment decoder is a logic circuit often used for the visual display of digital information. The seven outputs of the decoder will drive the seven segments on a corresponding display. BCD is the acronym for Binary Coded Decimal. The BCD system is used to represent the decimal numbers from 0 to 9 in a binary format suitable for digital devices. A four-bit code is required with the decimal characters 0 through 9 represented by the binary numbers 0000 through 1001. The combinations 1010 through 1111 are not used. A BCD to seven-segment decoder will allow the display of a binary coded decimal on a seven- segment display. The input to the decoder is a number from 0 through 9 in BCD and the output provides the seven inputs required to drive the seven-segment display.



Pin Names	Description
A0–A3	BCD Inputs
$\overline{RBI}$	Ripple Blanking Input (Active LOW)
$\overline{LT}$	Lamp Test Input (Active LOW)
$\overline{BI/RBO}$	Blanking Input (Active LOW) or Ripple Blanking Output (Active LOW)
$\overline{a}$ –$\overline{g}$	*Segment Outputs (Active LOW)

The seven segment display chip contains 7 LEDs, similar to the ones you have been using. The only difference is that they have been packaged into a single chip, and they have a different shape. The seven LEDs are not used individually; instead they are wired so their one end is common. If the anodes of the LEDs are wired together it is called “Common Anode” configuration and if the cathodes are connected together, it is known as “Common Cathode” configuration. Both the configurations are shown below.

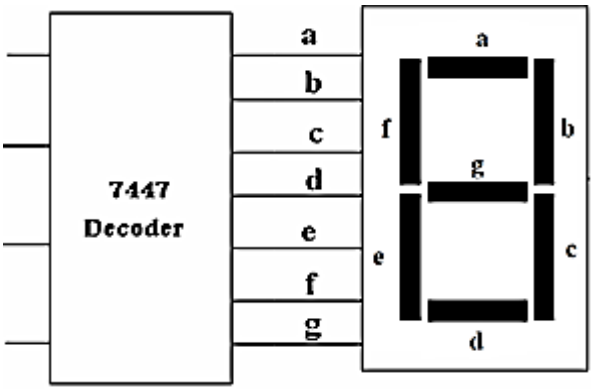




Task 1:

Steps:

1. Check the configuration of your 7-segment display using a Multi-meter. Find out its configuration and pin-assignment.
2. Create the truth table describing the function of a BCD to seven-segment decoder according to the configuration of your display. The lower case letters, a-g, represent the Segments on the display while the upper case letters A0-A3 represent the BCD input.
3. Design a circuit showing the proper connections of the decoder and the 7-segment display. Invert the outputs of the decoder in case of common cathode display. The output pins of 7447 are connected to the pins of seven segment display chip as shown below:

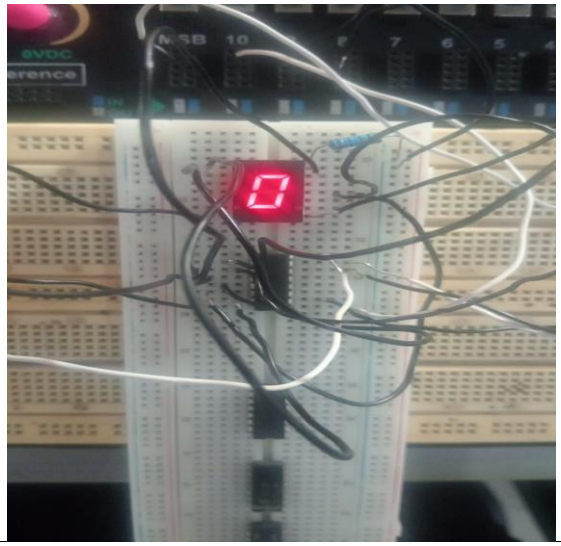


4. Implement the circuit on the trainer. Enter BCD numbers from 0 to 9 and see the corresponding decimal digit on the display.

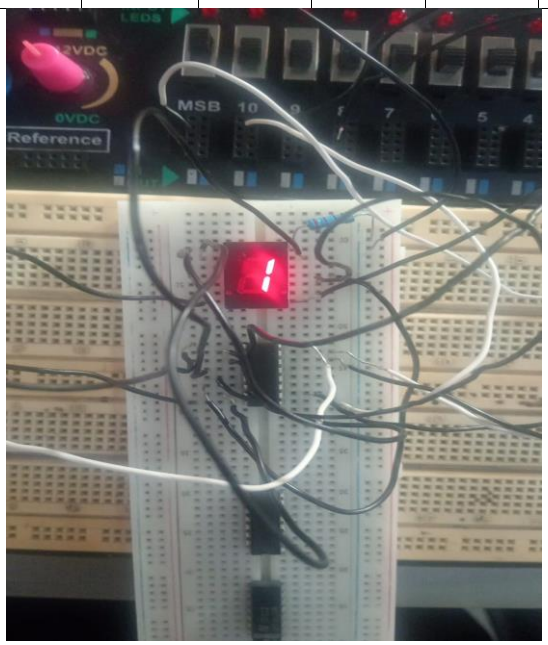
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**e**	**f**	**g**
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

Simulation Activity

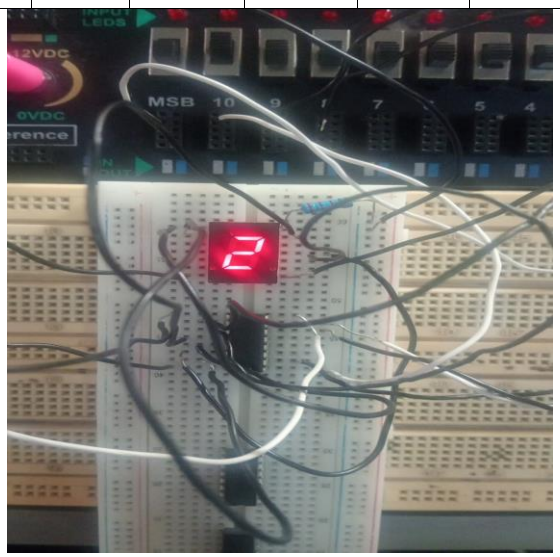
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**e**	**f**	**G**
0	0	0	0	1	1	1	1	1	1	0



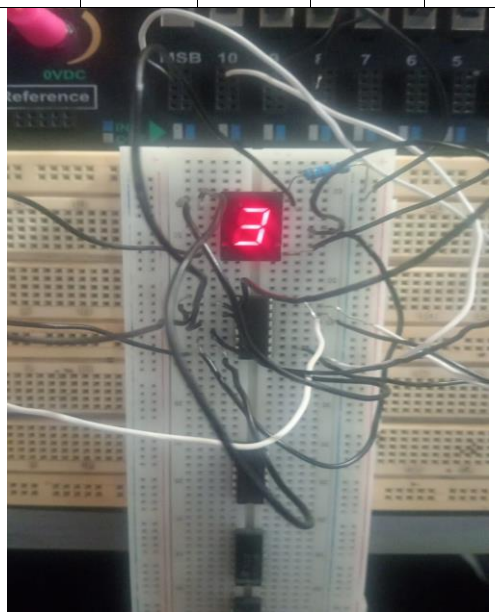
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**e**	**f**	**G**
0	0	0	1	0	1	1	0	0	0	0



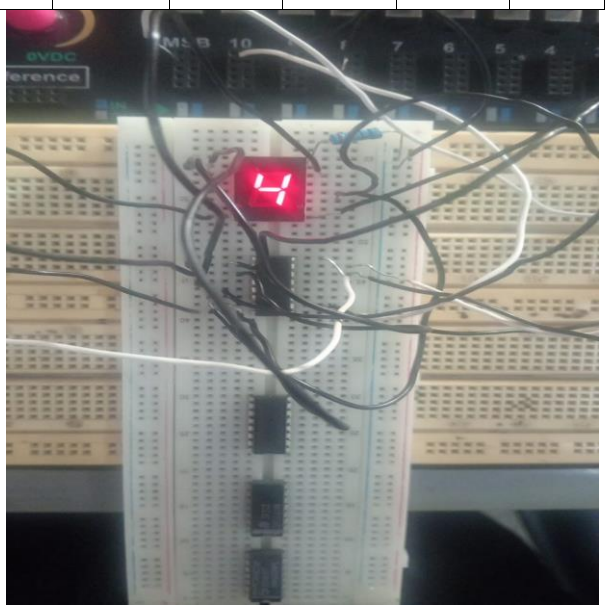
Inputs				Outputs						
A3	A2	Al	AO	a	b	c	d	e	f	G
0	0	1	0	1	1	0	1	1	0	1



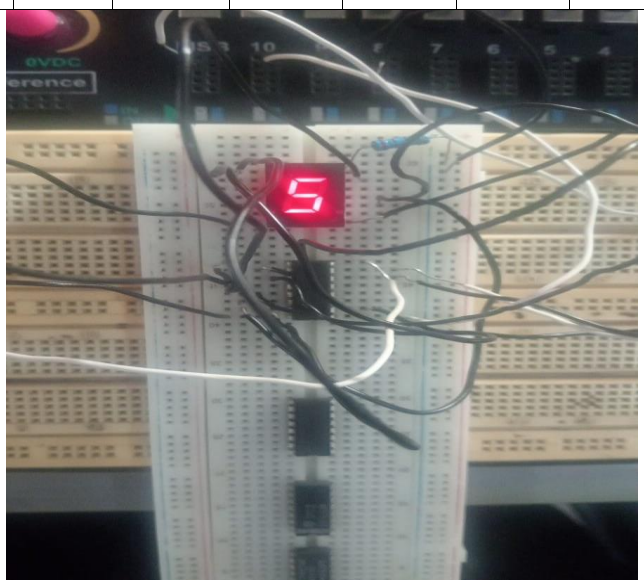
Inputs				Outputs						
A3	A2	Al	AO	a	b	c	D	e	f	G
0	0	1	1	1	1	1	1	0	0	1



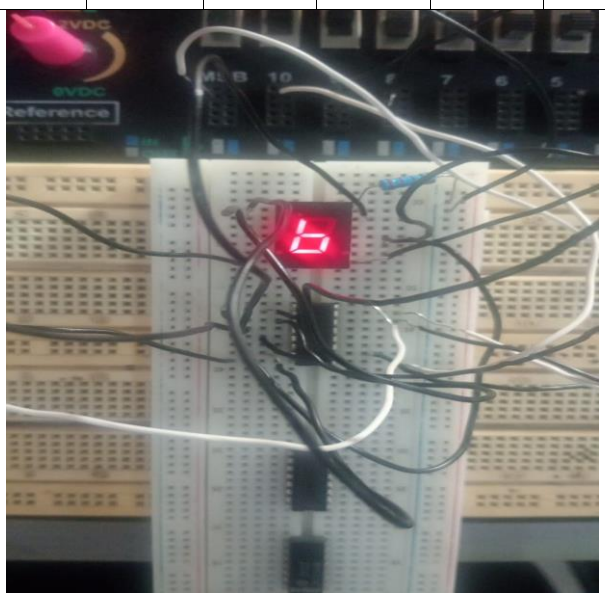
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**e**	**f**	**G**
0	1	0	0	0	1	1	0	0	1	1



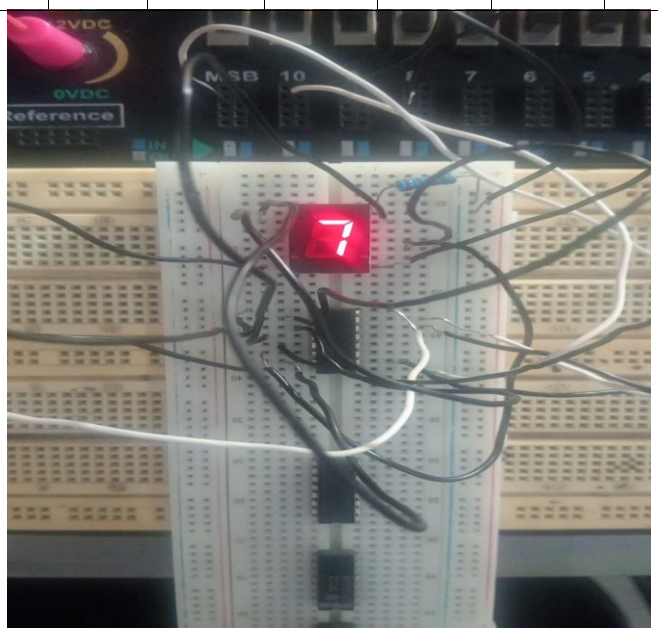
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**C**	**d**	**E**	**f**	**G**
0	1	0	1	1	0	1	1	0	1	1



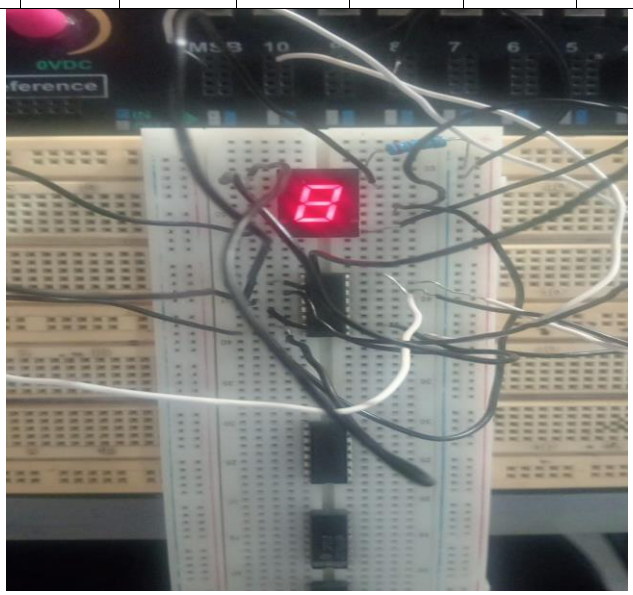
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**C**	**d**	**E**	**f**	**g**
0	1	1	0	1	0	1	1	1	1	1



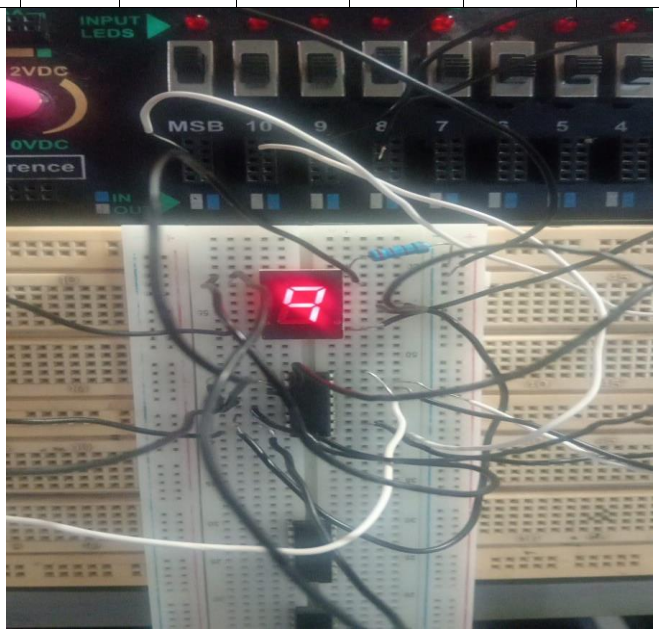
Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**E**	**f**	**g**
0	1	1	1	1	1	1	0	0	0	0



Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**D**	**E**	**f**	**g**
1	0	0	0	1	1	1	1	1	1	1



Inputs				Outputs						
A3	**A2**	**Al**	**AO**	**a**	**b**	**c**	**d**	**E**	**f**	**g**
1	0	0	1	1	1	1	1	0	1	1



Conclusion:

After performing this experiment, we successfully demonstrated the working of the BCD to 7-segment decoder (IC 7447) with a 7-segment display. The circuit correctly converted the Binary Coded Decimal (BCD) inputs into their corresponding decimal digits (0–9) on the display. Each 4-bit BCD input activated a unique combination of segments (a–g) to form the correct numeric character. This experiment helped us clearly understand how digital data in binary form can be visually represented as decimal numbers using a decoder and a display. It also showed how the 7447 IC simplifies the process of driving the 7-segment display by automatically decoding the BCD inputs. Overall, the experiment improved our understanding of number representation, digital-to-visual conversion, and the practical application of combinational logic circuits in real-world digital systems like calculators, counters, and electronic meters.